symcoder round-trip demo

electrons=(a,b,c), muons=(p,q)

1 Setup

Particle types: electrons = $\{a, b, c\}$, muons = $\{p, q\}$
Group G: $G = S_{\text{electrons}} \times S_{\text{muons}} = S_3 \times S_2$, order = 12

Operations:
$|\mathbf{x}|$ (rank 1, symmetric)
$\mathbf{x} \cdot \mathbf{y}$ (rank 2, symmetric)
$\mathbf{x}_T \cdot \mathbf{y}_T$ (rank 2, symmetric)
$\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (rank 3, antisymmetric)

Event vectors (3-D):
a: $\mathbf{a} = (1.20,\ 0.30,\ 0.50)$

b: $\mathbf{b} = (0.40,\ 1.10,\ 0.20)$

c: $\mathbf{c} = (-0.50,\ 1.20,\ -0.70)$

p: $\mathbf{p} = (0.00,\ 0.00,\ 1.00)$

q: $\mathbf{q} = (0.00,\ 0.00,\ -1.00)$

2 Phase 1 Encoding (one orbit per FlavouredOperator in repS)

11 FlavouredOperators in repS

2.1 FlavouredOperator: $|\mathbf{x}|$ flavour $= (1, 0)$

Canonical representative: $|\mathbf{a}|$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $|\mathbf{a}|$, $|\mathbf{b}|$, $|\mathbf{c}|$
Eval values: +1.3342, +1.1874, +1.4765
Encoded [3 reals]: +1.1874, +1.3342, +1.4765

2.2 FlavouredOperator: $|\mathbf{x}|$ flavour $= (0, 1)$

Canonical representative: $|\mathbf{p}|$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $|\mathbf{p}|$, $|\mathbf{q}|$

Eval values: +1.0000, +1.0000
Encoded [2 reals]: +1.0000, +1.0000

2.3 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (2, 0)$

Canonical representative: $\mathbf{a} \cdot \mathbf{b}$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$, $\mathbf{b} \cdot \mathbf{c}$
Eval values: +0.9100, -0.5900, +0.9800
Encoded [3 reals]: -0.5900, +0.9100, +0.9800

2.4 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (1, 1)$

Canonical representative: $\mathbf{a} \cdot \mathbf{p}$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{q}$
Eval values: +0.5000, -0.5000, +0.2000, -0.2000, -0.7000, +0.7000
Encoded [6 reals]: -0.7000, -0.5000, -0.2000, +0.2000, +0.5000, +0.7000

2.5 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour $= (0, 2)$

Canonical representative: $\mathbf{p} \cdot \mathbf{q}$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{p} \cdot \mathbf{q}$
Eval values: -1.0000
Encoded [1 reals]: -1.0000

2.6 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour $= (2, 0)$

Canonical representative: $\mathbf{a}_T \cdot \mathbf{b}_T$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a}_T \cdot \mathbf{b}_T$, $\mathbf{a}_T \cdot \mathbf{c}_T$, $\mathbf{b}_T \cdot \mathbf{c}_T$
Eval values: +0.8100, -0.2400, +1.1200
Encoded [3 reals]: -0.2400, +0.8100, +1.1200

2.7 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour $= (1, 1)$

Canonical representative: $\mathbf{a}_T \cdot \mathbf{p}_T$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{a}_T \cdot \mathbf{p}_T$, $\mathbf{a}_T \cdot \mathbf{q}_T$, $\mathbf{b}_T \cdot \mathbf{p}_T$, $\mathbf{b}_T \cdot \mathbf{q}_T$, $\mathbf{c}_T \cdot \mathbf{p}_T$, $\mathbf{c}_T \cdot \mathbf{q}_T$
Eval values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Encoded [6 reals]: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

2.8 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour $= (0, 2)$

Canonical representative: $\mathbf{p}_T \cdot \mathbf{q}_T$
Encoder: `SortEncoder` — stores eval of full orbit

Orbit atoms: $\mathbf{p}_T \cdot \mathbf{q}_T$
Eval values: +0.0000
Encoded [1 reals]: +0.0000

2.9 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (3, 0)$

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$
Encoder: `HalfSortEncoder` — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$
Eval values: -0.6430
Encoded [1 reals]: +0.6430

2.10 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (2, 1)$

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$
Encoder: `HalfSortEncoder` — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$
Eval values: +1.2000, -1.2000, +1.5900, -1.5900, +1.0300, -1.0300
Encoded [6 reals]: +1.0300, +1.0300, +1.2000, +1.2000, +1.5900, +1.5900

2.11 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour $= (1, 2)$

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$
Encoder: `HalfSortEncoder` — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$
Eval values: +0.0000, +0.0000, +0.0000
Encoded [3 reals]: +0.0000, +0.0000, +0.0000

3 Phase 2 Encoding (OverlapBlocks, complementarity-drop disabled)

3.1 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 2
Encoded values: +2.8284, +2.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ |\mathbf{q}|) \to (+1.0000,\ +1.0000)$
$(|\mathbf{q}|,\ |\mathbf{p}|) \to (+1.0000,\ +1.0000)$
PairFlavour overlap=(0, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(0, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(|\mathbf{p}|,\ |\mathbf{p}|) \to (+1.0000,\ +1.0000)$

$(|\mathbf{q}|,\ |\mathbf{q}|) \rightarrow (+1.0000,\ +1.0000)$

3.2 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +7.9962, -11.5993, +39.9808, -86.3973, +32.8452, -97.7244, -61.3879, -6.6542, -76.5400, +15.8009, -14.2916
Decoded pairs (u, v):
$(|\mathbf{p}|,\ |\mathbf{a}|) \rightarrow (+1.0000,\ +1.4765)$
$(|\mathbf{q}|,\ |\mathbf{a}|) \rightarrow (+1.0000,\ +1.4765)$
$(|\mathbf{p}|,\ |\mathbf{b}|) \rightarrow (+1.0000,\ +1.3342)$
$(|\mathbf{q}|,\ |\mathbf{b}|) \rightarrow (+1.0000,\ +1.3342)$
$(|\mathbf{p}|,\ |\mathbf{c}|) \rightarrow (+1.0000,\ +1.1874)$
$(|\mathbf{q}|,\ |\mathbf{c}|) \rightarrow (+1.0000,\ +1.1874)$

3.3 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.0000, +0.0000, +1.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{p}_T \cdot \mathbf{q}_T) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ \mathbf{p}_T \cdot \mathbf{q}_T) \rightarrow (+1.0000,\ +0.0000)$

3.4 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 4
Encoded values: +2.0000, -2.0000, +0.0000, -2.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{p} \cdot \mathbf{q}) \rightarrow (+1.0000,\ -1.0000)$
$(|\mathbf{q}|,\ \mathbf{p} \cdot \mathbf{q}) \rightarrow (+1.0000,\ -1.0000)$

3.5 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a}_T \cdot \mathbf{q}_T) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ \mathbf{a}_T \cdot \mathbf{p}_T) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{p}|,\ \mathbf{b}_T \cdot \mathbf{q}_T) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ \mathbf{b}_T \cdot \mathbf{p}_T) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{p}|,\ \mathbf{c}_T \cdot \mathbf{q}_T) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ \mathbf{c}_T \cdot \mathbf{p}_T) \rightarrow (+1.0000,\ +0.0000)$
PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000

Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a}_T \cdot \mathbf{p}_T) \to (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ \mathbf{a}_T \cdot \mathbf{q}_T) \to (+1.0000,\ +0.0000)$
$(|\mathbf{p}|,\ \mathbf{b}_T \cdot \mathbf{p}_T) \to (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ \mathbf{b}_T \cdot \mathbf{q}_T) \to (+1.0000,\ +0.0000)$
$(|\mathbf{p}|,\ \mathbf{c}_T \cdot \mathbf{p}_T) \to (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ \mathbf{c}_T \cdot \mathbf{q}_T) \to (+1.0000,\ +0.0000)$

3.6 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +0.0000, +15.7800, +0.0000, +23.1200, +0.0000, +19.8321, +0.0000, +9.4242, +0.0000, +1.9370, +0.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a} \cdot \mathbf{q}) \to (+1.0000,\ -0.7000)$
$(|\mathbf{q}|,\ \mathbf{a} \cdot \mathbf{p}) \to (+1.0000,\ -0.5000)$
$(|\mathbf{p}|,\ \mathbf{b} \cdot \mathbf{q}) \to (+1.0000,\ -0.2000)$
$(|\mathbf{q}|,\ \mathbf{b} \cdot \mathbf{p}) \to (+1.0000,\ +0.7000)$
$(|\mathbf{p}|,\ \mathbf{c} \cdot \mathbf{q}) \to (+1.0000,\ +0.2000)$
$(|\mathbf{q}|,\ \mathbf{c} \cdot \mathbf{p}) \to (+1.0000,\ +0.5000)$
PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +0.0000, +15.7800, +0.0000, +23.1200, +0.0000, +19.8321, +0.0000, +9.4242, +0.0000, +1.9370, +0.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a} \cdot \mathbf{p}) \to (+1.0000,\ -0.7000)$
$(|\mathbf{q}|,\ \mathbf{a} \cdot \mathbf{q}) \to (+1.0000,\ -0.5000)$
$(|\mathbf{p}|,\ \mathbf{b} \cdot \mathbf{p}) \to (+1.0000,\ -0.2000)$
$(|\mathbf{q}|,\ \mathbf{b} \cdot \mathbf{q}) \to (+1.0000,\ +0.7000)$
$(|\mathbf{p}|,\ \mathbf{c} \cdot \mathbf{p}) \to (+1.0000,\ +0.2000)$
$(|\mathbf{q}|,\ \mathbf{c} \cdot \mathbf{q}) \to (+1.0000,\ +0.5000)$

3.7 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +3.3800, +11.2559, +16.9000, +5.0236, +32.7347, -8.0034, +30.6042, -10.0540, +13.5109, -3.3303, +2.1214
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+1.0000,\ +1.1200)$
$(|\mathbf{q}|,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+1.0000,\ +1.1200)$
$(|\mathbf{p}|,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+1.0000,\ +0.8100)$
$(|\mathbf{q}|,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+1.0000,\ +0.8100)$
$(|\mathbf{p}|,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+1.0000,\ -0.2400)$
$(|\mathbf{q}|,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+1.0000,\ -0.2400)$

3.8 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +2.6000, +13.7566, +13.0000, +15.0264, +27.6329, +6.2214, +30.8987,

-1.6099, +18.1337, -1.8384, +4.4679
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \mathbf{a}\cdot\mathbf{b}) \rightarrow (+1.0000,\ -0.5900)$
$(|\mathbf{q}|,\ \mathbf{a}\cdot\mathbf{b}) \rightarrow (+1.0000,\ -0.5900)$
$(|\mathbf{p}|,\ \mathbf{a}\cdot\mathbf{c}) \rightarrow (+1.0000,\ +0.9800)$
$(|\mathbf{q}|,\ \mathbf{a}\cdot\mathbf{c}) \rightarrow (+1.0000,\ +0.9800)$
$(|\mathbf{p}|,\ \mathbf{b}\cdot\mathbf{c}) \rightarrow (+1.0000,\ +0.9100)$
$(|\mathbf{q}|,\ \mathbf{b}\cdot\mathbf{c}) \rightarrow (+1.0000,\ +0.9100)$

3.9 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q})) \rightarrow (+1.0000,\ +0.0000)$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q})) \rightarrow (+1.0000,\ -0.0000)$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q})) \rightarrow (+1.0000,\ -0.0000)$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q})) \rightarrow (+1.0000,\ -0.0000)$

3.10 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +12.0000, +76.0580, +320.5800, +988.6013, +2326.8903, +4270.6178, +6142.2191, +6877.2694, +5857.9958, +3630.0616, +1483.7636, +314.7569
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q})) \rightarrow (+1.0000,\ +1.5900)$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})) \rightarrow (+1.0000,\ +1.5900)$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q})) \rightarrow (+1.0000,\ +1.2000)$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p})) \rightarrow (+1.0000,\ +1.2000)$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q})) \rightarrow (+1.0000,\ +1.0300)$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p})) \rightarrow (+1.0000,\ +1.0300)$
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q})) \rightarrow (+1.0000,\ -1.5900)$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})) \rightarrow (+1.0000,\ -1.5900)$
... (4 more pairs) PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +12.0000, +76.0580, +320.5800, +988.6013, +2326.8903, +4270.6178, +6142.2191, +6877.2694, +5857.9958, +3630.0616, +1483.7636, +314.7569
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})) \rightarrow (+1.0000,\ +1.5900)$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q})) \rightarrow (+1.0000,\ +1.5900)$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p})) \rightarrow (+1.0000,\ +1.2000)$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q})) \rightarrow (+1.0000,\ +1.2000)$
$(|\mathbf{p}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p})) \rightarrow (+1.0000,\ +1.0300)$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q})) \rightarrow (+1.0000,\ +1.0300)$
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})) \rightarrow (+1.0000,\ -1.5900)$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q})) \rightarrow (+1.0000,\ -1.5900)$
... (4 more pairs)

3.11 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=4 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 4
Encoded values: +4.0000, +6.8269, +5.6538, +1.9978
Decoded pairs (u, v):
$(|\mathbf{p}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ -0.6430)$
$(|\mathbf{q}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ -0.6430)$
$(|\mathbf{p}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ +0.6430)$
$(|\mathbf{q}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.0000,\ +0.6430)$

3.12 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
Encoded values: +11.3083, +53.3240, +134.2109, +190.1584, +143.8079, +45.3503
Decoded pairs (u, v):
$(|\mathbf{a}|,\ |\mathbf{b}|) \to (+1.4765,\ +1.1874)$
$(|\mathbf{a}|,\ |\mathbf{c}|) \to (+1.4765,\ +1.3342)$
$(|\mathbf{b}|,\ |\mathbf{a}|) \to (+1.3342,\ +1.1874)$
$(|\mathbf{b}|,\ |\mathbf{c}|) \to (+1.3342,\ +1.4765)$
$(|\mathbf{c}|,\ |\mathbf{a}|) \to (+1.1874,\ +1.3342)$
$(|\mathbf{c}|,\ |\mathbf{b}|) \to (+1.1874,\ +1.4765)$
PairFlavour overlap=(1, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(|\mathbf{a}|,\ |\mathbf{a}|) \to (+1.1874,\ +1.1874)$
$(|\mathbf{b}|,\ |\mathbf{b}|) \to (+1.3342,\ +1.3342)$
$(|\mathbf{c}|,\ |\mathbf{c}|) \to (+1.4765,\ +1.4765)$

3.13 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: +3.9981, +0.0000, +5.3073, +0.0000, +2.3391, +0.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{p}_T \cdot \mathbf{q}_T) \to (+1.4765,\ +0.0000)$
$(|\mathbf{b}|,\ \mathbf{p}_T \cdot \mathbf{q}_T) \to (+1.1874,\ +0.0000)$
$(|\mathbf{c}|,\ \mathbf{p}_T \cdot \mathbf{q}_T) \to (+1.3342,\ +0.0000)$

3.14 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: +3.9981, -3.0000, +2.3073, -7.9962, -1.6590, -4.3073
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{p} \cdot \mathbf{q}) \to (+1.4765,\ -1.0000)$
$(|\mathbf{b}|,\ \mathbf{p} \cdot \mathbf{q}) \to (+1.3342,\ -1.0000)$
$(|\mathbf{c}|,\ \mathbf{p} \cdot \mathbf{q}) \to (+1.1874,\ -1.0000)$

3.15 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +15.9923, +0.0000, +117.1373, +0.0000, +519.6184, +0.0000, +1554.7706, +0.0000, +3305.7787, +0.0000, +5121.4826, +0.0000, +5825.2045, +0.0000, +4827.6985, +0.0000, +2843.1049, +0.0000, +1129.3678, +0.0000, +271.6938, +0.0000, +29.9358, +0.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{b}_T \cdot \mathbf{p}_T) \to (+1.4765,\ +0.0000)$
$(|\mathbf{a}|,\ \mathbf{b}_T \cdot \mathbf{q}_T) \to (+1.4765,\ +0.0000)$
$(|\mathbf{a}|,\ \mathbf{c}_T \cdot \mathbf{p}_T) \to (+1.4765,\ +0.0000)$
$(|\mathbf{a}|,\ \mathbf{c}_T \cdot \mathbf{q}_T) \to (+1.4765,\ +0.0000)$
$(|\mathbf{b}|,\ \mathbf{a}_T \cdot \mathbf{p}_T) \to (+1.3342,\ +0.0000)$
$(|\mathbf{b}|,\ \mathbf{a}_T \cdot \mathbf{q}_T) \to (+1.3342,\ +0.0000)$
$(|\mathbf{b}|,\ \mathbf{c}_T \cdot \mathbf{p}_T) \to (+1.3342,\ +0.0000)$
$(|\mathbf{b}|,\ \mathbf{c}_T \cdot \mathbf{q}_T) \to (+1.3342,\ +0.0000)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +7.9962, +0.0000, +26.5993, +0.0000, +47.1165, +0.0000, +46.8716, +0.0000, +24.8287, +0.0000, +5.4714, +0.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{a}_T \cdot \mathbf{p}_T) \to (+1.4765,\ +0.0000)$
$(|\mathbf{a}|,\ \mathbf{a}_T \cdot \mathbf{q}_T) \to (+1.4765,\ +0.0000)$
$(|\mathbf{b}|,\ \mathbf{b}_T \cdot \mathbf{p}_T) \to (+1.3342,\ +0.0000)$
$(|\mathbf{b}|,\ \mathbf{b}_T \cdot \mathbf{q}_T) \to (+1.3342,\ +0.0000)$
$(|\mathbf{c}|,\ \mathbf{c}_T \cdot \mathbf{p}_T) \to (+1.1874,\ +0.0000)$
$(|\mathbf{c}|,\ \mathbf{c}_T \cdot \mathbf{q}_T) \to (+1.1874,\ +0.0000)$

3.16 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +15.9923, +0.0000, +118.6973, +0.0000, +540.5384, +0.0000, +1681.8576, +0.0000, +3766.3481, +0.0000, +6224.2982, +0.0000, +7648.5883, +0.0000, +6936.4566, +0.0000, +4528.1048, +0.0000, +2020.0109, +0.0000, +553.0411, +0.0000, +70.2940, +0.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{b} \cdot \mathbf{p}) \to (+1.1874,\ -0.7000)$
$(|\mathbf{a}|,\ \mathbf{b} \cdot \mathbf{q}) \to (+1.3342,\ -0.7000)$
$(|\mathbf{a}|,\ \mathbf{c} \cdot \mathbf{p}) \to (+1.4765,\ -0.5000)$
$(|\mathbf{a}|,\ \mathbf{c} \cdot \mathbf{q}) \to (+1.1874,\ -0.5000)$
$(|\mathbf{b}|,\ \mathbf{a} \cdot \mathbf{p}) \to (+1.3342,\ +0.7000)$
$(|\mathbf{b}|,\ \mathbf{a} \cdot \mathbf{q}) \to (+1.1874,\ +0.7000)$
$(|\mathbf{b}|,\ \mathbf{c} \cdot \mathbf{p}) \to (+1.4765,\ +0.5000)$
$(|\mathbf{b}|,\ \mathbf{c} \cdot \mathbf{q}) \to (+1.4765,\ -0.2000)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +7.9962, +0.0000, +27.3793, +0.0000, +51.1445, +0.0000, +54.8162, +0.0000, +31.8945, +0.0000, +7.8591, +0.0000
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \mathbf{a} \cdot \mathbf{p}) \to (+1.4765,\ -0.7000)$
$(|\mathbf{a}|,\ \mathbf{a} \cdot \mathbf{q}) \to (+1.4765,\ +0.7000)$
$(|\mathbf{b}|,\ \mathbf{b} \cdot \mathbf{p}) \to (+1.3342,\ +0.5000)$
$(|\mathbf{b}|,\ \mathbf{b} \cdot \mathbf{q}) \to (+1.1874,\ +0.2000)$

$(|\mathbf{c}|,\ \mathbf{c}\cdot\mathbf{p})\rightarrow(+1.3342,\ -0.5000)$
$(|\mathbf{c}|,\ \mathbf{c}\cdot\mathbf{q})\rightarrow(+1.1874,\ -0.2000)$

3.17 Block: $|\mathbf{x}|\times\mathbf{x}_T\cdot\mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0,0)$, type=11, kind=ASSOC, encoded dim= 6
Encoded values: +3.9981, +1.6900, +4.8633, +4.3515, +1.9181, +2.9918
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \mathbf{b}_T\cdot\mathbf{c}_T)\rightarrow(+1.3342,\ +1.1200)$
$(|\mathbf{b}|,\ \mathbf{a}_T\cdot\mathbf{c}_T)\rightarrow(+1.4765,\ +0.8100)$
$(|\mathbf{c}|,\ \mathbf{a}_T\cdot\mathbf{b}_T)\rightarrow(+1.1874,\ -0.2400)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1,0)$, type=11, kind=ASSOC, encoded dim= 12
Encoded values: +7.9962, +3.3800, +22.8552, +22.6755, +26.7283, +59.6993, +4.7645, +76.6858, -14.1741, +47.4925, -8.2128, +11.0421
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \mathbf{a}_T\cdot\mathbf{b}_T)\rightarrow(+1.4765,\ +1.1200)$
$(|\mathbf{a}|,\ \mathbf{a}_T\cdot\mathbf{c}_T)\rightarrow(+1.1874,\ +1.1200)$
$(|\mathbf{b}|,\ \mathbf{a}_T\cdot\mathbf{b}_T)\rightarrow(+1.1874,\ +0.8100)$
$(|\mathbf{b}|,\ \mathbf{b}_T\cdot\mathbf{c}_T)\rightarrow(+1.3342,\ +0.8100)$
$(|\mathbf{c}|,\ \mathbf{a}_T\cdot\mathbf{c}_T)\rightarrow(+1.4765,\ -0.2400)$
$(|\mathbf{c}|,\ \mathbf{b}_T\cdot\mathbf{c}_T)\rightarrow(+1.3342,\ -0.2400)$

3.18 Block: $|\mathbf{x}|\times\mathbf{x}\cdot\mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0,0)$, type=11, kind=ASSOC, encoded dim= 6
Encoded values: +3.9981, +1.3000, +5.5306, +3.2470, +2.8502, +2.5238
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \mathbf{b}\cdot\mathbf{c})\rightarrow(+1.4765,\ +0.9100)$
$(|\mathbf{b}|,\ \mathbf{a}\cdot\mathbf{c})\rightarrow(+1.3342,\ +0.9800)$
$(|\mathbf{c}|,\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.1874,\ -0.5900)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1,0)$, type=11, kind=ASSOC, encoded dim= 12
Encoded values: +7.9962, +2.6000, +25.3559, +17.5430, +39.9914, +48.9122, +30.3174, +69.9165, +6.7367, +51.0380, -2.4527, +15.0917
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.4765,\ -0.5900)$
$(|\mathbf{a}|,\ \mathbf{a}\cdot\mathbf{c})\rightarrow(+1.3342,\ -0.5900)$
$(|\mathbf{b}|,\ \mathbf{a}\cdot\mathbf{b})\rightarrow(+1.4765,\ +0.9800)$
$(|\mathbf{b}|,\ \mathbf{b}\cdot\mathbf{c})\rightarrow(+1.3342,\ +0.9100)$
$(|\mathbf{c}|,\ \mathbf{a}\cdot\mathbf{c})\rightarrow(+1.1874,\ +0.9100)$
$(|\mathbf{c}|,\ \mathbf{b}\cdot\mathbf{c})\rightarrow(+1.1874,\ +0.9800)$

3.19 Block: $|\mathbf{x}|\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=12, kind=ASSOC, encoded dim= 12
Encoded values: +15.9923, +117.1373, +519.6184, +1554.7706, +3305.7787, +5121.4826, +5825.2045, +4827.6985, +2843.1049, +1129.3678, +271.6938, +29.9358
Decoded pairs (u,v):

$(|\mathbf{a}|,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.4765, +0.0000)
$(|\mathbf{a}|,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.4765, +0.0000)
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.4765, +0.0000)
$(|\mathbf{b}|,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.4765, −0.0000)
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.3342, −0.0000)
$(|\mathbf{c}|,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.3342, −0.0000)
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.3342, +0.0000)
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.3342, +0.0000)
... (4 more pairs) PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(1,0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: +7.9962, +26.5993, +47.1165, +46.8716, +24.8287, +5.4714
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.4765, +0.0000)
$(|\mathbf{b}|,\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.4765, +0.0000)
$(|\mathbf{c}|,\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.3342, +0.0000)
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.3342, −0.0000)
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.1874, −0.0000)
$(|\mathbf{c}|,\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow$ (+1.1874, −0.0000)

3.20 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +15.9923, +127.1953, +654.2952, +2406.8990, +6640.4046, +14054.4075, +22965.6441, +28758.7978, +26951.0762, +17999.4163, +7735.4211, +1640.2216
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow$ (+1.1874, +1.5900)
$(|\mathbf{a}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow$ (+1.1874, +1.5900)
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow$ (+1.4765, +1.2000)
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow$ (+1.4765, +1.2000)
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow$ (+1.3342, +1.0300)
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow$ (+1.3342, +1.0300)
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow$ (+1.1874, −1.5900)
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow$ (+1.1874, −1.5900)
... (4 more pairs) PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +15.9923, +0.0000, +127.1953, +0.0000, +653.3436, +0.0000, +2395.3663, +0.0000, +6572.5381, -0.0000, +13804.9212, -0.0000, +22339.0427, -0.0000, +27649.1661, -0.0000, +25568.2570, +0.0000, +16830.6354, -0.0000, +7126.1835, -0.0000, +1490.5721, -0.0000
Decoded pairs (u,v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow$ (+1.4765, −1.5900)
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow$ (+1.3342, −1.5900)
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow$ (+1.4765, −1.0300)
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow$ (+1.3342, −1.2000)
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow$ (+1.1874, −1.2000)
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow$ (+1.1874, −1.0300)
$(|\mathbf{b}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow$ (+1.4765, +1.5900)
$(|\mathbf{b}|,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow$ (+1.3342, +1.5900)
... (4 more pairs)

3.21 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: +7.9962, +27.8397, +53.7285, +60.6021, +37.9391, +10.3729
Decoded pairs (u, v):
$(|\mathbf{a}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.4765,\ +0.6430)$
$(|\mathbf{b}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.3342,\ +0.6430)$
$(|\mathbf{c}|,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.1874,\ +0.6430)$
$(|\mathbf{a}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.4765,\ -0.6430)$
$(|\mathbf{b}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.3342,\ -0.6430)$
$(|\mathbf{c}|,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+1.1874,\ -0.6430)$

3.22 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(0, 2)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{p}_T \cdot \mathbf{q}_T) \to (+0.0000,\ +0.0000)$

3.23 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 2) type=11 kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 2)$, type=`11`, kind=`ASSOC`, encoded dim= 2
Encoded values: +0.0000, -1.0000
Decoded pairs (u, v):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{p} \cdot \mathbf{q}) \to (+0.0000,\ -1.0000)$

3.24 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{p}_T) \to (+0.0000,\ +0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{q}_T) \to (+0.0000,\ +0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{p}_T) \to (+0.0000,\ +0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{q}_T) \to (+0.0000,\ +0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{p}_T) \to (+0.0000,\ +0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{q}_T) \to (+0.0000,\ +0.0000)$

3.25 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.0000, +0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, +0.0049, +0.0000
Decoded pairs (u, v):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{p}) \to (+0.0000,\ -0.7000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{q}) \to (+0.0000,\ -0.5000)$

$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{p}) \to (+0.0000,\ +0.7000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{q}) \to (+0.0000,\ +0.5000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{p}) \to (+0.0000,\ -0.2000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{q}) \to (+0.0000,\ +0.2000)$

3.26 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: +0.0000, +1.6900, -0.4440, +0.0000, +0.0000, +0.2177
Decoded pairs (u, v):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.0000,\ +1.1200)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+0.0000,\ +0.8100)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+0.0000,\ -0.2400)$

3.27 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: +0.0000, +1.3000, +0.2233, +0.0000, +0.0000, +0.5262
Decoded pairs (u, v):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+0.0000,\ -0.5900)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{c}) \to (+0.0000,\ +0.9800)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{c}) \to (+0.0000,\ +0.9100)$

3.28 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 2)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$

3.29 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163
Decoded pairs (u, v):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (+0.0000,\ -1.5900)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (+0.0000,\ -1.5900)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (+0.0000,\ -1.2000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (+0.0000,\ -1.2000)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (+0.0000,\ -1.0300)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (+0.0000,\ -1.0300)$

$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (+0.0000,\ +1.5900)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \rightarrow (+0.0000,\ +1.5900)$
... (4 more pairs)

3.30 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 2
Encoded values: +0.0000, +0.4134
Decoded pairs (u, v):
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \rightarrow (+0.0000,\ -0.6430)$
$(\mathbf{p}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \rightarrow (+0.0000,\ +0.6430)$

3.31 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(0, 2)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{p} \cdot \mathbf{q}) \rightarrow (-1.0000,\ -1.0000)$

3.32 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: -6.0000, +0.0000, +15.0000, +0.0000, -20.0000, +0.0000, +15.0000, +0.0000, -6.0000, +0.0000, +1.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{p}_T) \rightarrow (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{q}_T) \rightarrow (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b}_T \cdot \mathbf{p}_T) \rightarrow (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b}_T \cdot \mathbf{q}_T) \rightarrow (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c}_T \cdot \mathbf{p}_T) \rightarrow (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c}_T \cdot \mathbf{q}_T) \rightarrow (-1.0000,\ +0.0000)$

3.33 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: -6.0000, +0.0000, +15.7800, +0.0000, -23.1200, +0.0000, +19.8321, +0.0000, -9.4242, +0.0000, +1.9370, +0.0000
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{p}) \rightarrow (-1.0000,\ +0.7000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{q}) \rightarrow (-1.0000,\ +0.5000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{p}) \rightarrow (-1.0000,\ +0.2000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{q}) \rightarrow (-1.0000,\ -0.7000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{p}) \rightarrow (-1.0000,\ -0.2000)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{q}) \rightarrow (-1.0000,\ -0.5000)$

3.34 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: -3.0000, +1.6900, +2.5560, -3.3800, -0.5560, +1.9077
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (-1.0000,\ +1.1200)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (-1.0000,\ +0.8100)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (-1.0000,\ -0.2400)$

3.35 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: -3.0000, +1.3000, +3.2233, -2.6000, -1.2233, +1.8262
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{b}) \to (-1.0000,\ -0.5900)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{c}) \to (-1.0000,\ +0.9800)$
$(\mathbf{p} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{c}) \to (-1.0000,\ +0.9100)$

3.36 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 2)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: -6.0000, +15.0000, -20.0000, +15.0000, -6.0000, +1.0000
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ +0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ -0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ -0.0000)$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (-1.0000,\ -0.0000)$

3.37 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: -12.0000, +76.0580, -320.5800, +988.6013, -2326.8903, +4270.6178, -6142.2191, +6877.2694, -5857.9958, +3630.0616, -1483.7636, +314.7569
Decoded pairs (u, v):
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-1.0000,\ -1.5900)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (-1.0000,\ -1.5900)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (-1.0000,\ -1.2000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (-1.0000,\ -1.2000)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (-1.0000,\ -1.0300)$
$(\mathbf{p} \cdot \mathbf{q},\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (-1.0000,\ -1.0300)$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (-1.0000,\ +1.5900)$
$(\mathbf{p} \cdot \mathbf{q},\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (-1.0000,\ +1.5900)$
... (4 more pairs)

3.38 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= $(0,0)$, type=`12`, kind=`ASSOC`, encoded dim= 2
Encoded values: -2.0000, +1.4134
Decoded pairs (u,v):
$(\mathbf{p}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(-1.0000,\ -0.6430)$
$(\mathbf{p}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(-1.0000,\ +0.6430)$

3.39 Block: $\mathbf{x}_T\cdot\mathbf{y}_T\times\mathbf{x}_T\cdot\mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \mathbf{b}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \mathbf{b}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \mathbf{c}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \mathbf{c}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \mathbf{a}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ \mathbf{a}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \mathbf{c}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ \mathbf{c}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
... (4 more pairs) PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0,1)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \mathbf{b}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \mathbf{b}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \mathbf{c}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \mathbf{c}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \mathbf{a}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ \mathbf{a}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \mathbf{c}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ \mathbf{c}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1,0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \mathbf{a}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \mathbf{a}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \mathbf{b}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ \mathbf{b}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{c}_T\cdot\mathbf{p}_T,\ \mathbf{c}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{c}_T\cdot\mathbf{q}_T,\ \mathbf{c}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1,1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u,v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \mathbf{a}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \mathbf{a}_T\cdot\mathbf{q}_T)\to(+0.0000,\ +0.0000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \mathbf{b}_T\cdot\mathbf{p}_T)\to(+0.0000,\ +0.0000)$

$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{q}_T)\rightarrow$ (+0.0000, +0.0000)
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{c}_T \cdot \mathbf{p}_T)\rightarrow$ (+0.0000, +0.0000)
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{c}_T \cdot \mathbf{q}_T)\rightarrow$ (+0.0000, +0.0000)

3.40 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +1.5600, +0.0000, +0.0000, +0.0000, +0.9126, +0.0000, +0.0000, +0.0000, +0.2471, +0.0000, +0.0000, +0.0000, +0.0308, +0.0000, +0.0000, +0.0000, +0.0015, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{q})\rightarrow$ (+0.0000, +0.7000)
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{p})\rightarrow$ (+0.0000, +0.7000)
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{q})\rightarrow$ (+0.0000, −0.7000)
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{p})\rightarrow$ (+0.0000, −0.7000)
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{q})\rightarrow$ (+0.0000, −0.5000)
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{p})\rightarrow$ (+0.0000, −0.5000)
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{q})\rightarrow$ (+0.0000, +0.5000)
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{p})\rightarrow$ (+0.0000, +0.5000)
... (4 more pairs) PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +1.5600, +0.0000, +0.0000, +0.0000, +0.9126, +0.0000, +0.0000, +0.0000, +0.2471, +0.0000, +0.0000, +0.0000, +0.0308, +0.0000, +0.0000, +0.0000, +0.0015, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{p})\rightarrow$ (+0.0000, +0.7000)
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{q})\rightarrow$ (+0.0000, +0.7000)
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{p})\rightarrow$ (+0.0000, −0.7000)
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{q})\rightarrow$ (+0.0000, −0.7000)
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{p})\rightarrow$ (+0.0000, −0.5000)
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{q})\rightarrow$ (+0.0000, −0.5000)
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{p})\rightarrow$ (+0.0000, +0.5000)
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{q})\rightarrow$ (+0.0000, +0.5000)
... (4 more pairs) PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.0000, +0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, +0.0049, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{q})\rightarrow$ (+0.0000, −0.7000)
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{p})\rightarrow$ (+0.0000, −0.5000)
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{q})\rightarrow$ (+0.0000, +0.7000)
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{p})\rightarrow$ (+0.0000, +0.5000)
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{q})\rightarrow$ (+0.0000, −0.2000)
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{p})\rightarrow$ (+0.0000, +0.2000)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.0000, +0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, +0.0049, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{p})\rightarrow$ (+0.0000, −0.7000)
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{q})\rightarrow$ (+0.0000, −0.5000)
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{p})\rightarrow$ (+0.0000, +0.7000)

$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{q}) \to (+0.0000,\ +0.5000)$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{c} \cdot \mathbf{p}) \to (+0.0000,\ -0.2000)$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{c} \cdot \mathbf{q}) \to (+0.0000,\ +0.2000)$

3.41 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +3.3800, -3.7441, +0.0000, +0.0000, -1.0653, -0.5388, +0.0000, +0.0000, -0.1933, -0.0474, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+0.0000,\ +1.1200)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+0.0000,\ +1.1200)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+0.0000,\ +0.8100)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+0.0000,\ +0.8100)$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.0000,\ -0.2400)$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.0000,\ -0.2400)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +6.7600, -18.9126, +0.0000, +0.0000, -27.4406, +20.1419, +0.0000, +0.0000, +3.9480, +4.1119, +0.0000, +0.0000, +2.2752, +0.2333, +0.0000, +0.0000, +0.3093, +0.0137, +0.0000, +0.0000, +0.0183, +0.0022, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.0000,\ +1.1200)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.0000,\ +1.1200)$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+0.0000,\ +1.1200)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+0.0000,\ +1.1200)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.0000,\ +0.8100)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.0000,\ +0.8100)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+0.0000,\ +0.8100)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+0.0000,\ +0.8100)$
... (4 more pairs)

3.42 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +2.6000, -1.2434, +0.0000, +0.0000, +1.6329, -1.3182, +0.0000, +0.0000, +0.2350, -0.2768, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{c}) \to (+0.0000,\ -0.5900)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{c}) \to (+0.0000,\ -0.5900)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{c}) \to (+0.0000,\ +0.9800)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{c}) \to (+0.0000,\ +0.9800)$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+0.0000,\ +0.9100)$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+0.0000,\ +0.9100)$
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +5.2000, -9.2468, +0.0000, +0.0000, -3.1999, -9.5814, +0.0000, +0.0000, -10.4452, -1.1640, +0.0000, +0.0000, -6.3288, +1.6586, +0.0000, +0.0000, -1.5236, +0.6746, +0.0000, +0.0000, -0.1301, +0.0766, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+0.0000,\ -0.5900)$

$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+0.0000,\ -0.5900)$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{c}) \to (+0.0000,\ -0.5900)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{c}) \to (+0.0000,\ -0.5900)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+0.0000,\ +0.9800)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+0.0000,\ +0.9800)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \mathbf{b} \cdot \mathbf{c}) \to (+0.0000,\ +0.9800)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \mathbf{b} \cdot \mathbf{c}) \to (+0.0000,\ +0.9800)$
... (4 more pairs)

3.43 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ +0.0000)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+0.0000,\ -0.0000)$

3.44 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (+0.0000,\ -1.5900)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (+0.0000,\ -1.5900)$
$(\mathbf{b}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \to (+0.0000,\ -1.2000)$
$(\mathbf{b}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \to (+0.0000,\ -1.2000)$
$(\mathbf{c}_T \cdot \mathbf{p}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \to (+0.0000,\ -1.0300)$
$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \to (+0.0000,\ -1.0300)$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \to (+0.0000,\ +1.5900)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \to (+0.0000,\ +1.5900)$
... (4 more pairs) PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap=

$(0,1)$, type=12, kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163
Decoded pairs (u,v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ -1.5900)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+0.0000,\ -1.5900)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ -1.2000)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(+0.0000,\ -1.2000)$
$(\mathbf{c}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.0000,\ -1.0300)$
$(\mathbf{c}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.0000,\ -1.0300)$
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ +1.5900)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+0.0000,\ +1.5900)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=11, kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +10.0580, +0.0000, +0.0000, +0.0000, +40.9913, +0.0000, +0.0000, +0.0000, +86.6819, +0.0000, +0.0000, +0.0000, +100.4717, +0.0000, +0.0000, +0.0000, +60.6378, +0.0000, +0.0000, +0.0000, +14.9163, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.0000,\ -1.5900)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.0000,\ -1.5900)$
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(+0.0000,\ -1.2000)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ -1.2000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.0000,\ -1.0300)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.0000,\ -1.0300)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+0.0000,\ +1.5900)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ +1.5900)$
... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,1)$, type=11, kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +10.0580, +0.0000, +0.0000, +0.0000, +40.9913, +0.0000, +0.0000, +0.0000, +86.6819, +0.0000, +0.0000, +0.0000, +100.4717, +0.0000, +0.0000, +0.0000, +60.6378, +0.0000, +0.0000, +0.0000, +14.9163, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.0000,\ -1.5900)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.0000,\ -1.5900)$
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ -1.2000)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(+0.0000,\ -1.2000)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.0000,\ -1.0300)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.0000,\ -1.0300)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+0.0000,\ +1.5900)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+0.0000,\ +1.5900)$
... (4 more pairs)

3.45 Block: $\mathbf{x}_T\cdot\mathbf{y}_T\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1,0)$, type=12, kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +2.4807, +0.0000, +2.5641, +0.0000, +1.4135, +0.0000, +0.4383, +0.0000, +0.0725, +0.0000, +0.0050
Decoded pairs (u,v):
$(\mathbf{a}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -0.6430)$
$(\mathbf{a}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -0.6430)$
$(\mathbf{b}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -0.6430)$
$(\mathbf{b}_T\cdot\mathbf{q}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -0.6430)$
$(\mathbf{c}_T\cdot\mathbf{p}_T,\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -0.6430)$

$(\mathbf{c}_T \cdot \mathbf{q}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+0.0000,\ -0.6430)$
$(\mathbf{a}_T \cdot \mathbf{p}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+0.0000,\ +0.6430)$
$(\mathbf{a}_T \cdot \mathbf{q}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \to (+0.0000,\ +0.6430)$
... (4 more pairs)

3.46 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, -1.5600, +0.0000, +0.9126, -0.0000, -0.2198, +0.0000, +0.0095, +0.0000, +0.0027, +0.0000, +0.0129
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{b} \cdot \mathbf{q}) \to (+0.5000,\ +0.7000)$
$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{p}) \to (+0.7000,\ +0.5000)$
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{c} \cdot \mathbf{q}) \to (+0.7000,\ +0.2000)$
$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{p}) \to (+0.2000,\ +0.7000)$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{q}) \to (+0.5000,\ -0.2000)$
$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{p}) \to (+0.2000,\ -0.5000)$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{c} \cdot \mathbf{q}) \to (-0.2000,\ -0.7000)$
$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{p}) \to (-0.5000,\ -0.7000)$
... (4 more pairs) PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +1.5600, -0.0000, +0.9126, -0.0000, +0.2198, -0.0000, +0.0095, -0.0000, -0.0027, +0.0000, +0.0129
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{b} \cdot \mathbf{p}) \to (+0.7000,\ -0.5000)$
$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{q}) \to (+0.5000,\ -0.7000)$
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{c} \cdot \mathbf{p}) \to (+0.2000,\ -0.7000)$
$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{q}) \to (+0.7000,\ -0.2000)$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{p}) \to (-0.2000,\ -0.5000)$
$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{q}) \to (-0.5000,\ -0.2000)$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{c} \cdot \mathbf{p}) \to (-0.7000,\ +0.2000)$
$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{q}) \to (-0.7000,\ +0.5000)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
Encoded values: +0.0000, +1.5600, +0.0000, +0.6084, +0.0000, +0.0392
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{q}) \to (+0.7000,\ -0.7000)$
$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{p}) \to (+0.5000,\ -0.5000)$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{b} \cdot \mathbf{q}) \to (-0.7000,\ +0.7000)$
$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{p}) \to (-0.5000,\ +0.5000)$
$(\mathbf{c} \cdot \mathbf{p},\ \mathbf{c} \cdot \mathbf{q}) \to (+0.2000,\ -0.2000)$
$(\mathbf{c} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{p}) \to (-0.2000,\ +0.2000)$
PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{p}) \to (-0.7000,\ -0.7000)$
$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{q}) \to (-0.5000,\ -0.5000)$
$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{b} \cdot \mathbf{p}) \to (-0.2000,\ -0.2000)$
$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{q}) \to (+0.2000,\ +0.2000)$
$(\mathbf{c} \cdot \mathbf{p},\ \mathbf{c} \cdot \mathbf{p}) \to (+0.5000,\ +0.5000)$
$(\mathbf{c} \cdot \mathbf{q},\ \mathbf{c} \cdot \mathbf{q}) \to (+0.7000,\ +0.7000)$

3.47 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=`11`, kind=`ASSOC`, encoded dim= 12

Encoded values: +0.0000, +3.3800, -4.5241, -0.0000, +0.0000, -2.3671, -0.0651, +0.0000, -0.0000, -0.3392, -0.1683, +0.0000

Decoded pairs (u, v):

$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+0.5000,\ +1.1200)$

$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+0.7000,\ +0.8100)$

$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (-0.5000,\ +1.1200)$

$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (-0.7000,\ +0.8100)$

$(\mathbf{c} \cdot \mathbf{p},\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.2000,\ -0.2400)$

$(\mathbf{c} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (-0.2000,\ -0.2400)$

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 24

Encoded values: +0.0000, +6.7600, -20.4726, +0.0000, -0.0000, -36.6844, +44.4512, -0.0000, +0.0000, +41.4253, -33.6972, -0.0000, +0.0000, -24.2761, +14.0721, +0.0000, -0.0000, +6.4766, -2.8173, -0.0000, +0.0000, -1.1257, +0.2399, +0.0000

Decoded pairs (u, v):

$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (-0.7000,\ -0.2400)$

$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (-0.7000,\ +1.1200)$

$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (-0.5000,\ -0.2400)$

$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+0.7000,\ -0.2400)$

$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.5000,\ -0.2400)$

$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+0.7000,\ +1.1200)$

$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (-0.5000,\ +0.8100)$

$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (-0.2000,\ +1.1200)$

... (4 more pairs)

3.48 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=`11`, kind=`ASSOC`, encoded dim= 12

Encoded values: +0.0000, +2.6000, -2.0234, +0.0000, +0.0000, +0.9395, -1.6868, +0.0000, +0.0000, -0.0250, -0.6192, +0.0000

Decoded pairs (u, v):

$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{b} \cdot \mathbf{c}) \to (-0.7000,\ +0.9100)$

$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{b} \cdot \mathbf{c}) \to (-0.5000,\ +0.9800)$

$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{c}) \to (+0.7000,\ +0.9100)$

$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{c}) \to (+0.5000,\ +0.9800)$

$(\mathbf{c} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{b}) \to (-0.2000,\ -0.5900)$

$(\mathbf{c} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{b}) \to (+0.2000,\ -0.5900)$

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 24

Encoded values: +0.0000, +5.2000, -10.8068, -0.0000, +0.0000, -10.6185, +5.1708, +0.0000, -0.0000, +5.4759, -11.3264, +0.0000, +0.0000, -10.0996, +2.7457, +0.0000, +0.0000, +0.6239, -2.8179, +0.0000, -0.0000, -2.5284, +0.6807, -0.0000

Decoded pairs (u, v):

$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{b}) \to (-0.7000,\ -0.5900)$

$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{b}) \to (-0.5000,\ -0.5900)$

$(\mathbf{a} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{c}) \to (+0.7000,\ -0.5900)$

$(\mathbf{a} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{c}) \to (+0.5000,\ -0.5900)$

$(\mathbf{b} \cdot \mathbf{p},\ \mathbf{a} \cdot \mathbf{b}) \to (-0.7000,\ +0.9800)$

$(\mathbf{b} \cdot \mathbf{q},\ \mathbf{a} \cdot \mathbf{b}) \to (+0.7000,\ +0.9800)$

$(\mathbf{b}\cdot\mathbf{p},\ \mathbf{b}\cdot\mathbf{c})\to(-0.5000,\ +0.9100)$
$(\mathbf{b}\cdot\mathbf{q},\ \mathbf{b}\cdot\mathbf{c})\to(+0.5000,\ +0.9100)$
... (4 more pairs)

3.49 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(0,1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, -1.5600, +0.0000, +0.0000, +0.0000, +0.9126, +0.0000, +0.0000, +0.0000, -0.2471, +0.0000, -0.0000, +0.0000, +0.0308, +0.0000, +0.0000, +0.0000, -0.0015, +0.0000, -0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(+0.7000,\ +0.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(+0.7000,\ +0.0000)$
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(-0.7000,\ +0.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(-0.7000,\ -0.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.5000,\ -0.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.5000,\ -0.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.5000,\ +0.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.5000,\ +0.0000)$
... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1,1)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.0000, -0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, -0.0049, +0.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.7000,\ +0.0000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.5000,\ +0.0000)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(+0.7000,\ +0.0000)$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(+0.5000,\ -0.0000)$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(-0.2000,\ -0.0000)$
$(\mathbf{c}\cdot\mathbf{q},\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.2000,\ -0.0000)$

3.50 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +8.4980, +0.0000, +32.6451, +0.0000, +74.9539, +0.0000, +108.9323, +0.0000, +93.2282, +0.0000, +42.2161
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+0.2000,\ -1.5900)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(-0.2000,\ -1.5900)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to(+0.7000,\ -1.2000)$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to(+0.5000,\ -1.0300)$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(-0.7000,\ -1.2000)$
$(\mathbf{c}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(-0.5000,\ -1.0300)$
$(\mathbf{a}\cdot\mathbf{p},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+0.7000,\ +1.2000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+0.2000,\ +1.5900)$
... (4 more pairs) PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0,1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +8.4980, +0.0000, +32.6451, +0.0000, +74.9539, +0.0000, +108.9323, +0.0000, +93.2282, +0.0000, +42.2161
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+0.2000,\ -1.5900)$

$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(-0.2000,\ -1.5900)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(+0.7000,\ -1.2000)$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(+0.5000,\ -1.0300)$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-0.7000,\ -1.2000)$
$(\mathbf{c}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-0.5000,\ -1.0300)$
$(\mathbf{a}\cdot\mathbf{p},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+0.7000,\ +1.2000)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+0.2000,\ +1.5900)$
... (4 more pairs) PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +8.4980, +6.3900, -0.0000, -0.0000, +14.3494, +44.1434, -0.0000, +0.0000, -28.3068, +104.9126, -0.0000, -0.0000, -107.5651, +100.7828, +0.0000, -0.0000, -108.4640, +22.6706, +0.0000, -0.0000, -33.8292, -11.7340
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.7000,\ -1.5900)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.5000,\ -1.5900)$
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(-0.2000,\ -1.2000)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(+0.7000,\ -1.0300)$
$(\mathbf{b}\cdot\mathbf{p},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.5000,\ -1.2000)$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.2000,\ -1.0300)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(-0.7000,\ +1.5900)$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(-0.5000,\ +1.5900)$
... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +8.4980, -6.3900, +0.0000, +0.0000, +14.3494, -44.1434, +0.0000, -0.0000, -28.3068, -104.9126, +0.0000, +0.0000, -107.5651, -100.7828, +0.0000, +0.0000, -108.4640, -22.6706, +0.0000, -0.0000, -33.8292, +11.7340
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.7000,\ +1.5900)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.5000,\ +1.5900)$
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(-0.2000,\ +1.2000)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(+0.7000,\ +1.0300)$
$(\mathbf{b}\cdot\mathbf{p},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.5000,\ +1.2000)$
$(\mathbf{b}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.2000,\ +1.0300)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(-0.7000,\ -1.5900)$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(-0.5000,\ -1.5900)$
... (4 more pairs)

3.51 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1,0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +0.9207, -0.0000, +1.5418, -0.0000, +1.1362, -0.0000, +0.7563, -0.0000, +0.3093, -0.0000, +0.0739
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.7000,\ -0.6430)$
$(\mathbf{a}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.7000,\ -0.6430)$
$(\mathbf{b}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.5000,\ -0.6430)$
$(\mathbf{b}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.5000,\ -0.6430)$
$(\mathbf{c}\cdot\mathbf{p},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.2000,\ -0.6430)$
$(\mathbf{c}\cdot\mathbf{q},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.2000,\ -0.6430)$
$(\mathbf{a}\cdot\mathbf{p},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.7000,\ +0.6430)$
$(\mathbf{a}\cdot\mathbf{q},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.7000,\ +0.6430)$
... (4 more pairs)

3.52 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
Encoded values: +4.7800, +10.5364, +14.1307, +12.0340, +6.2732, +1.7889
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (-0.2400,\ +1.1200)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (-0.2400,\ +0.8100)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+0.8100,\ +1.1200)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (+1.1200,\ +0.8100)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+1.1200,\ -0.2400)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+0.8100,\ -0.2400)$
PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{a}_T \cdot \mathbf{b}_T) \to (-0.2400,\ -0.2400)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{a}_T \cdot \mathbf{c}_T) \to (+0.8100,\ +0.8100)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{b}_T \cdot \mathbf{c}_T) \to (+1.1200,\ +1.1200)$

3.53 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 12
Encoded values: +3.3800, +2.6000, +2.5007, +8.5673, -4.5116, +11.3877, -9.5180, +5.4343, -6.3584, -0.9672, -1.2926, -1.7926
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{a} \cdot \mathbf{c}) \to (+1.1200,\ -0.5900)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+0.8100,\ -0.5900)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{b} \cdot \mathbf{c}) \to (+1.1200,\ +0.9100)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+0.8100,\ +0.9800)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{b} \cdot \mathbf{c}) \to (-0.2400,\ +0.9800)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{a} \cdot \mathbf{c}) \to (-0.2400,\ +0.9100)$
PairFlavour overlap=(2, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= $(2, 0)$, type=`11`, kind=`ASSOC`, encoded dim= 6
Encoded values: +1.6900, +1.3000, +0.6673, +0.2207, +1.0660, -0.4442
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \mathbf{a} \cdot \mathbf{b}) \to (+1.1200,\ +0.9800)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \mathbf{a} \cdot \mathbf{c}) \to (+0.8100,\ +0.9100)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \mathbf{b} \cdot \mathbf{c}) \to (-0.2400,\ -0.5900)$

3.54 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: +3.3800, +3.7441, +1.0653, -0.5388, -0.1933, +0.0474
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+1.1200,\ +0.0000)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (+1.1200,\ +0.0000)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (+0.8100,\ +0.0000)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \to (+0.8100,\ -0.0000)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \to (-0.2400,\ -0.0000)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \to (-0.2400,\ -0.0000)$
PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`12`,

kind=ASSOC, encoded dim= 12
Encoded values: +6.7600, +18.9126, +27.4406, +20.1419, +3.9480, -4.1119, -2.2752, +0.2333, +0.3093, -0.0137, -0.0183, +0.0022
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \rightarrow (+1.1200,\ +0.0000)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \rightarrow (+1.1200,\ +0.0000)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \rightarrow (+1.1200,\ +0.0000)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})) \rightarrow (+1.1200,\ -0.0000)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \rightarrow (+0.8100,\ -0.0000)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})) \rightarrow (+0.8100,\ -0.0000)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \rightarrow (+0.8100,\ +0.0000)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})) \rightarrow (+0.8100,\ +0.0000)$
... (4 more pairs)

3.55 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 0)$, type=11, kind=ASSOC, encoded dim= 24
Encoded values: +6.7600, +0.0000, +28.9706, +0.0000, +81.9304, -0.0000, +182.5098, -0.0000, +317.3109, -0.0000, +471.8350, -0.0000, +574.4634, +0.0000, +612.5140, +0.0000, +514.8609, +0.0000, +384.0874, +0.0000, +198.4450, +0.0000, +93.3348, +0.0000
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \rightarrow (+1.1200,\ -1.5900)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \rightarrow (+0.8100,\ -1.5900)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (-0.2400,\ -1.2000)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \rightarrow (-0.2400,\ -1.0300)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \rightarrow (+1.1200,\ -1.2000)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \rightarrow (+0.8100,\ -1.0300)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (+1.1200,\ +1.5900)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \rightarrow (+0.8100,\ +1.5900)$
... (4 more pairs) PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(2, 0)$, type=12, kind=ASSOC, encoded dim= 12
Encoded values: +6.7600, +28.9706, +88.4413, +218.6210, +440.5627, +747.9918, +1056.1305, +1254.6568, +1207.9684, +922.4338, +489.6353, +157.4689
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (-0.2400,\ -1.5900)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \rightarrow (-0.2400,\ -1.5900)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})) \rightarrow (+1.1200,\ -1.0300)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})) \rightarrow (+1.1200,\ -1.0300)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})) \rightarrow (+0.8100,\ -1.2000)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})) \rightarrow (+0.8100,\ -1.2000)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})) \rightarrow (-0.2400,\ +1.5900)$
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})) \rightarrow (-0.2400,\ +1.5900)$
... (4 more pairs)

3.56 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(2, 0)$, type=12, kind=ASSOC, encoded dim= 6
Encoded values: +3.3800, +4.9844, +3.8602, +2.3357, +1.5450, +0.8403
Decoded pairs (u, v):
$(\mathbf{a}_T \cdot \mathbf{b}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \rightarrow (-0.2400,\ -0.6430)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \rightarrow (+1.1200,\ -0.6430)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})) \rightarrow (+0.8100,\ -0.6430)$

$(\mathbf{a}_T \cdot \mathbf{b}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c})) \rightarrow (+1.1200,\ +0.6430)$
$(\mathbf{a}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c})) \rightarrow (+0.8100,\ +0.6430)$
$(\mathbf{b}_T \cdot \mathbf{c}_T,\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c})) \rightarrow (-0.2400,\ +0.6430)$

3.57 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`11_sigma`, kind=`ASSOC`, encoded dim= 6
Encoded values: +3.6770, +7.2066, +10.3451, +9.7461, +6.3330, +2.7526
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \mathbf{a} \cdot \mathbf{c}) \rightarrow (+0.9800,\ -0.5900)$
$(\mathbf{a} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{b}) \rightarrow (+0.9100,\ -0.5900)$
$(\mathbf{a} \cdot \mathbf{b},\ \mathbf{b} \cdot \mathbf{c}) \rightarrow (+0.9800,\ +0.9100)$
$(\mathbf{b} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{b}) \rightarrow (+0.9100,\ +0.9800)$
$(\mathbf{a} \cdot \mathbf{c},\ \mathbf{b} \cdot \mathbf{c}) \rightarrow (-0.5900,\ +0.9800)$
$(\mathbf{b} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{c}) \rightarrow (-0.5900,\ +0.9100)$
PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2, 0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \mathbf{a} \cdot \mathbf{b}) \rightarrow (-0.5900,\ -0.5900)$
$(\mathbf{a} \cdot \mathbf{c},\ \mathbf{a} \cdot \mathbf{c}) \rightarrow (+0.9100,\ +0.9100)$
$(\mathbf{b} \cdot \mathbf{c},\ \mathbf{b} \cdot \mathbf{c}) \rightarrow (+0.9800,\ +0.9800)$

3.58 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(0, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 6
Encoded values: +2.6000, +1.2434, -1.6329, -1.3182, +0.2350, +0.2768
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q})) \rightarrow (-0.5900,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q})) \rightarrow (-0.5900,\ +0.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})) \rightarrow (+0.9800,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q})) \rightarrow (+0.9800,\ -0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q})) \rightarrow (+0.9100,\ -0.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})) \rightarrow (+0.9100,\ -0.0000)$
PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(1, 0)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +5.2000, +9.2468, +3.1999, -9.5814, -10.4452, +1.1640, +6.3288, +1.6586, -1.5236, -0.6746, +0.1301, +0.0766
Decoded pairs (u, v):
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})) \rightarrow (-0.5900,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})) \rightarrow (-0.5900,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q})) \rightarrow (-0.5900,\ +0.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q})) \rightarrow (-0.5900,\ -0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q})) \rightarrow (+0.9800,\ -0.0000)$
$(\mathbf{b} \cdot \mathbf{c},\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q})) \rightarrow (+0.9800,\ -0.0000)$
$(\mathbf{a} \cdot \mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})) \rightarrow (+0.9800,\ +0.0000)$
$(\mathbf{a} \cdot \mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})) \rightarrow (+0.9800,\ +0.0000)$
... (4 more pairs)

3.59 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=11, kind=ASSOC, encoded dim= 24
Encoded values: +5.2000, +0.0000, +19.3048, -0.0000, +44.1431, -0.0000, +88.3113, -0.0000, +137.8122, -0.0000, +217.1767, -0.0000, +269.5218, -0.0000, +324.6213, -0.0000, +280.9677, -0.0000, +279.5834, -0.0000, +185.4809, -0.0000, +133.7534, -0.0000
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(-0.5900,\ +1.2000)$
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(-0.5900,\ +1.0300)$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.9800,\ +1.5900)$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.9100,\ +1.5900)$
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+0.9800,\ +1.2000)$
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+0.9100,\ +1.0300)$
$(\mathbf{b}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-0.5900,\ -1.2000)$
$(\mathbf{b}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-0.5900,\ -1.0300)$
... (4 more pairs) PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(2,0)$, type=12, kind=ASSOC, encoded dim= 12
Encoded values: +5.2000, +19.3048, +52.0675, +125.1057, +249.4131, +435.0611, +636.0779, +824.2870, +888.3968, +786.4776, +473.5668, +173.8698
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-0.5900,\ -1.5900)$
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-0.5900,\ -1.5900)$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(-0.5900,\ +1.5900)$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(-0.5900,\ +1.5900)$
$(\mathbf{b}\cdot\mathbf{c},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+0.9100,\ -1.2000)$
$(\mathbf{b}\cdot\mathbf{c},\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+0.9100,\ -1.2000)$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+0.9800,\ -1.0300)$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+0.9800,\ -1.0300)$
... (4 more pairs)

3.60 Block: $\mathbf{x}\cdot\mathbf{y}\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= $(2,0)$, type=12, kind=ASSOC, encoded dim= 6
Encoded values: +2.6000, +2.4837, +0.5170, +0.5921, +1.7446, +1.2990
Decoded pairs (u,v):
$(\mathbf{a}\cdot\mathbf{b},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.5900,\ -0.6430)$
$(\mathbf{a}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.5900,\ +0.6430)$
$(\mathbf{b}\cdot\mathbf{c},\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.9800,\ -0.6430)$
$(\mathbf{a}\cdot\mathbf{b},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.9100,\ -0.6430)$
$(\mathbf{a}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.9800,\ +0.6430)$
$(\mathbf{b}\cdot\mathbf{c},\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.9100,\ +0.6430)$

3.61 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=neg_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(0,2)$, type=neg_sigma, kind=ASSOC, encoded dim= 6
Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow(+0.0000,\ +0.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\rightarrow(+0.0000,\ +0.0000)$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(+0.0000,\ +0.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\rightarrow(-0.0000,\ -0.0000)$

$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
... (4 more pairs) PairFlavour overlap=(1, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(1, 2)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(+0.0000,\ +0.0000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$
$(-\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}))\to(-0.0000,\ -0.0000)$

3.62 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= $(0, 1)$, type=`12`, kind=`ASSOC`, encoded dim= 12
Encoded values: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+0.0000,\ -1.5900)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+0.0000,\ -1.5900)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to(+0.0000,\ -1.2000)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to(-0.0000,\ -1.2000)$
$(\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(-0.0000,\ -1.0300)$
$(-\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(-0.0000,\ -1.0300)$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+0.0000,\ +1.5900)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+0.0000,\ +1.5900)$
... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1, 1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +10.0580, +0.0000, +0.0000, +0.0000, +40.9913, +0.0000, +0.0000, +0.0000, +86.6819, +0.0000, +0.0000, +0.0000, +100.4717, +0.0000, +0.0000, +0.0000, +60.6378, +0.0000, +0.0000, +0.0000, +14.9163, +0.0000
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(+0.0000,\ -1.5900)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(+0.0000,\ -1.5900)$
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to(+0.0000,\ -1.2000)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to(-0.0000,\ -1.2000)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(-0.0000,\ -1.0300)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(-0.0000,\ -1.0300)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+0.0000,\ +1.5900)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+0.0000,\ +1.5900)$
... (4 more pairs)

3.63 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=22 kind=ASSOC dim=6 **PairFlavour** overlap= $(1, 0)$, type=`22`, kind=`ASSOC`, encoded dim= 6
Encoded values: -2.4807, +2.5641, -1.4135, +0.4383, -0.0725, +0.0050
Decoded pairs (u, v):
$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+0.0000,\ +0.6430)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+0.0000,\ +0.6430)$

$(\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -0.6430)$
$(-\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.0000,\ -0.6430)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.0000,\ +0.6430)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(-0.0000,\ +0.6430)$
$(\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -0.6430)$
$(-\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\rightarrow(+0.0000,\ -0.6430)$
... (4 more pairs)

3.64 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,0)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +0.0000, +9.2388, -0.0000, -0.0000, -14.7572, +0.0000, +0.0000, -0.0000, +0.0000, +98.5481, -0.0000, +0.0000, -301.4282, +0.0000, +0.0000, -0.0000, -0.0000, +257.2546, +0.0000, +0.0000, -1268.5418, -0.0000
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(-1.0300,\ +1.5900)$
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(-1.2000,\ +1.5900)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-1.5900,\ +1.2000)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-1.5900,\ +1.0300)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+1.0300,\ +1.2000)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+1.2000,\ +1.0300)$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-1.2000,\ -1.0300)$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-1.0300,\ -1.2000)$
... (4 more pairs) PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= $(1,1)$, type=`11`, kind=`ASSOC`, encoded dim= 24
Encoded values: +0.0000, +0.0000, +0.0000, -9.2388, +0.0000, -0.0000, -14.7572, -0.0000, +0.0000, +0.0000, -0.0000, -98.5481, +0.0000, -0.0000, -301.4282, -0.0000, -0.0000, -0.0000, -0.0000, -257.2546, +0.0000, +0.0000, -1268.5418, +0.0000
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(+1.5900,\ +1.0300)$
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(+1.5900,\ +1.2000)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+1.2000,\ +1.5900)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+1.0300,\ +1.5900)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\rightarrow(+1.2000,\ -1.0300)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\rightarrow(+1.0300,\ -1.2000)$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-1.0300,\ +1.2000)$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-1.2000,\ +1.0300)$
... (4 more pairs) PairFlavour overlap=(2, 0) type=neg_sigma kind=ASSOC dim=6 **PairFlavour** overlap= $(2,0)$, type=`neg_sigma`, kind=`ASSOC`, encoded dim= 6
Encoded values: -20.1160, -163.9651, +693.4549, +1607.5466, -1940.4080, -954.6460
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(+1.5900,\ -1.5900)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\rightarrow(-1.5900,\ +1.5900)$
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+1.5900,\ -1.5900)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(-1.5900,\ +1.5900)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(+1.2000,\ -1.2000)$
$(-\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\rightarrow(-1.2000,\ +1.2000)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(+1.2000,\ -1.2000)$
$(-\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\rightarrow(-1.2000,\ +1.2000)$
... (4 more pairs) PairFlavour overlap=(2, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(2,1)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\rightarrow(+1.0300,\ +1.0300)$

$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(+1.0300,\ +1.0300)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}))\to(+1.2000,\ +1.2000)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}))\to(+1.2000,\ +1.2000)$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{p}))\to(+1.5900,\ +1.5900)$
$(\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{b},\mathbf{c},\mathbf{q}))\to(+1.5900,\ +1.5900)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}))\to(-1.0300,\ -1.0300)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}))\to(-1.0300,\ -1.0300)$
... (4 more pairs)

3.65 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=neg kind=ASSOC dim=12 **PairFlavour** overlap= $(2,0)$, type=`neg`, kind=`ASSOC`, encoded dim= 12
Encoded values: +7.5773, +0.0000, +31.0800, -0.0000, +72.8464, -0.0000, +116.6920, -0.0000, +108.6872, -0.0000, +64.6121, -0.0000
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+1.5900,\ -0.6430)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(-1.5900,\ +0.6430)$
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+1.5900,\ +0.6430)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(-1.5900,\ -0.6430)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+1.2000,\ -0.6430)$
$(-\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(-1.2000,\ +0.6430)$
$(\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+1.0300,\ -0.6430)$
$(-\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(-1.0300,\ +0.6430)$
... (4 more pairs)

3.66 Block: $\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})\times\varepsilon(\mathbf{x},\mathbf{y},\mathbf{z})$ (block)

PairFlavour overlap=(3, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= $(3,0)$, type=`null_self`, kind=`NULL_SELF`, encoded dim= 0
Decoded pairs (u,v):
$(\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}),\ \varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(+0.6430,\ +0.6430)$
$(-\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}),\ -\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c}))\to(-0.6430,\ -0.6430)$

4 Alignment Decoder (recovers the G-orbit of repS evaluations)

—**repS**—: $|repS| = 11$

—**G**—: $|G| = 12$

Decoded vectors: decoded column vectors: 12

Decoded alignment table (rows=repS atoms, cols=group elements): **Decoded alignment table** (rows= repS atoms, columns= $g\in G$):

Ground truth (direct evaluation of g·repS on event): **Ground truth** (direct evaluation of $g\cdot$repS on event):

Multiset comparison (decoded vs ground truth): **Multiset comparison:**
MISMATCH — decoded multiset differs from ground truth
Max —error— per repS row (decoded vs ground truth): **Max |error| per repS row:**
Mymag(a): $|\mathbf{a}|$: max err $= 0.00e+00$

Table 1: Decoded alignment table

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	
$\lvert\mathbf{a}\rvert$	+1.4765	+1.4765	+1.3342	+1.3342	+1.1874	+1.1874	+1.1874	+1.1874	+1.3342	+1.3
$\lvert\mathbf{p}\rvert$	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0
$\mathbf{a} \cdot \mathbf{b}$	-0.5900	+0.9800	-0.5900	+0.9100	+0.9800	+0.9100	+0.9100	+0.9800	+0.9100	-0.5
$\mathbf{a} \cdot \mathbf{p}$	-0.7000	-0.7000	-0.5000	-0.5000	-0.2000	-0.2000	+0.2000	+0.2000	+0.5000	+0.5
$\mathbf{p} \cdot \mathbf{q}$	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0
$\mathbf{a}_T \cdot \mathbf{b}_T$	-0.2400	+1.1200	-0.2400	+0.8100	+1.1200	+0.8100	+0.8100	+1.1200	+0.8100	-0.2
$\mathbf{a}_T \cdot \mathbf{p}_T$	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0
$\mathbf{p}_T \cdot \mathbf{q}_T$	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$	-1.5900	-1.0300	-1.5900	-1.2000	-1.0300	+1.2000	-1.2000	+1.0300	+1.2000	+1.5
$\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$	+0.0000	+0.0000	+0.0000	+0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0

Table 2: Ground truth alignment table

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	
$\lvert\mathbf{a}\rvert$	+1.3342	+1.3342	+1.3342	+1.3342	+1.1874	+1.1874	+1.1874	+1.1874	+1.4765	+1.4
$\lvert\mathbf{p}\rvert$	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0
$\mathbf{a} \cdot \mathbf{b}$	+0.9100	+0.9100	-0.5900	-0.5900	+0.9100	+0.9100	+0.9800	+0.9800	-0.5900	-0.5
$\mathbf{a} \cdot \mathbf{p}$	+0.5000	-0.5000	+0.5000	-0.5000	+0.2000	-0.2000	+0.2000	-0.2000	-0.7000	+0.7
$\mathbf{p} \cdot \mathbf{q}$	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0
$\mathbf{a}_T \cdot \mathbf{b}_T$	+0.8100	+0.8100	-0.2400	-0.2400	+0.8100	+0.8100	+1.1200	+1.1200	-0.2400	-0.2
$\mathbf{a}_T \cdot \mathbf{p}_T$	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0
$\mathbf{p}_T \cdot \mathbf{q}_T$	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$	-0.6430	-0.6430	+0.6430	+0.6430	+0.6430	+0.6430	-0.6430	-0.6430	-0.6430	-0.6
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$	+1.2000	-1.2000	+1.5900	-1.5900	-1.2000	+1.2000	+1.0300	-1.0300	-1.5900	+1.5
$\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0

Mymag(p): $|\mathbf{p}|$: max err $= 0.00e + 00$

dot(a,b): $\mathbf{a} \cdot \mathbf{b}$: max err $= 0.00e + 00$

dot(a,p): $\mathbf{a} \cdot \mathbf{p}$: max err $= 0.00e + 00$

dot(p,q): $\mathbf{p} \cdot \mathbf{q}$: max err $= 0.00e + 00$

dot(a,b): $\mathbf{a}_T \cdot \mathbf{b}_T$: max err $= 0.00e + 00$

dot(a,p): $\mathbf{a}_T \cdot \mathbf{p}_T$: max err $= 0.00e + 00$

dot(p,q): $\mathbf{p}_T \cdot \mathbf{q}_T$: max err $= 0.00e + 00$

eps(a,b,c): $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$: max err $= 1.29e + 00$

eps(a,b,p): $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$: max err $= 0.00e + 00$

eps(a,p,q): $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$: max err $= 0.00e + 00$

5 Summary

Phase 1 encoded length: Phase 1 encoded length: 35 reals

Phase 2 encoded length: Phase 2 encoded length: 1214 reals

Total encoded length: Total encoded length: 1249 reals

repS size: |repS| = 11 atoms

—G—: $|G| = 12$ group elements

Decoded vectors: decoded alignment vectors: 12

TeX output written to: /Users/lester/github/Echinocoder/symcoder/demo_roundtrip.tex Compile with: pdflatex demo_roundtrip.tex